

Residential Solar Panel Installation Guide

GreenGrid Energy Solutions

January 2026

1 Introduction

This guide provides homeowners with a practical overview of residential solar panel installation, including system components, installation steps, cost considerations, and expected financial returns.

2 System Overview

A typical residential solar power system consists of:

- Photovoltaic (PV) panels
- Inverter (string or micro-inverter)
- Mounting and racking system
- Electrical wiring and safety disconnects
- Grid connection and net metering interface

3 Site Requirements

- Roof condition: structurally sound with at least 20 years lifespan remaining
- Roof orientation: south-facing preferred (east/west acceptable)
- Roof tilt: 20–40 degrees optimal
- Minimal shading between 9:00–16:00

4 Installation Process

1. Site inspection and energy assessment
2. System design and permitting
3. Mounting structure installation
4. Panel and inverter installation
5. Electrical connection and testing
6. Utility inspection and grid approval

5 Cost Breakdown

Component	Estimated Cost (USD)
Solar Panels (6 kW system)	9,000
Inverter	2,000
Mounting & Hardware	1,200
Labor & Installation	3,500
Permits & Inspection	800
Total Installed Cost	16,500

6 Energy Production

A 6 kW residential system typically produces:

- Annual output: 7,500–9,000 kWh
- Performance degradation: 0.5% per year

7 Financial Benefits

- Reduced electricity bills
- Net metering credits
- Government tax credits or rebates (region dependent)

8 Maintenance

- Annual inspection recommended
- Panel cleaning: 1–2 times per year
- Inverter replacement after 10–12 years

9 Safety and Compliance

All installations must comply with local electrical codes and utility interconnection standards.

Document Version: 1.3

Prepared by: Engineering Department, GreenGrid Energy Solutions